IRJTAE

Ingenious Research Journal for Technological Advancements in Engineering

(Open Access, Peer-Reviewed, Technological Journal)

Volume:03/Issue:01/March-2026 **www.irjtae.com**

FARMNEXIS +: A SMART AGRICULTURE PLATFORM LEVERAGING BLOCKCHAIN

Khushi S. Burbure*1, Likhita G. Parise*2, Falguni K. Barlawar*3, Sharvari A. Shirbhate*4, Pooja S. Biradar*5, Shrut S. Raut*6, Prof. A.V. Mahalle*7

*1 to 6 Students, Department of Computer Engineering, Jagadambha College of Engineering and Technology Yavatmal, Maharashtra, India.

*7Assistant Professor, Department of Computer Engineering, Jagadambha College of Engineering and Technology Yavatmal, Maharashtra, India.

ABSTRACT

The research emphasizes real-time interactions through social media supported by Blockchain-enabled mechanisms such as immutable ledgers and smart contracts, which provide reliable and verifiable feedback. Addressing challenges, limitations, and ethical considerations, this study asserts that the fusion of Blockchain and social media strengthens trust, accountability, and long-term relationships between organizations and stakeholders. Our research utilizes digital traces of opinions, interactions, and experiences generated on social media platforms. By integrating Blockchain architecture with a well-established social media framework, we construct a robust and transparent analytical foundation. Experimental observations indicate that Blockchain-based frameworks significantly improve data security, authenticity, and stakeholder confidence when compared with traditional centralized systems. Furthermore, Blockchain facilitates secure identity management, transparent engagement tracking, and decentralized content validation, thereby enhancing stakeholder participation and organizational credibility in digital environments.

Keywords: Blockchain Technology, Social Media Platforms, Stakeholder Engagement, Decentralization, Data Security, Transparency, Digital Communication, Smart Contracts, Data Integrity, Trust Management.

I INTRODUCTION

Developed using HTML, CSS, and JavaScript for the frontend to create cross-platform user interfaces, the system utilizes FastAPI for AI model deployment and Spring Boot for backend coordination and service management. The platform is supported by cloud infrastructure and integrated databases, including MySQL for structured data and NoSQL databases for handling unstructured and large-scale data. FarmNexis+ aims to bridge the digital divide in agriculture, promote nutritional awareness, and establish trust and transparency in the agri-food supply chain, representing a significant step toward sustainable and technology-driven farming.

II.METHODOLOGY

www.irjtae.com

Blockchain integration is achieved using the Polygon network, where smart contracts record every transaction and supply-chain event in an immutable ledger, ensuring transparency and traceability. For financial transactions,

the Unified Payments Interface (UPI) is incorporated to enable direct, fast, and secure payments between farmers and consumers.

1. System Design and Architecture

The proposed smart agriculture platform is designed by integrating Blockchain technology and data management systems. The system aims to improve transparency, data security, and automation in agricultural processes.

- The platform consists of the following main components:
- Phones to collect photos of yield from farmer only one option to the farmer upload from camera will provide to maintain the truest of consumer**1**
- Blockchain network for secure data storage
- Smart contracts for automation
- User interface for farmers and stakeholders

2. Blockchain Network Setup

The platform uses a permissioned or public blockchain such as:

- Ethereum
- Hyperledger Fabric
- Blockchain functions:
- Stores sensor data securely

3. System Architecture Development

- The FarmNexis platform is designed using blockchain technology integrated with cloud-based data storage. Smart contracts were developed to automate transactions such as crop data recording, supply chain tracking, and payments.
- Finally, the entire platform is deployed using cloud technologies such as Firebase and Render, with Docker used for containerization. This deployment strategy ensures scalability, reliability, and ease of maintenance across different environments.

III. MODELLING AND ANALYSIS

The FarmNexis+ system is modelled as a layered smart agriculture platform integrating Artificial Intelligence, Blockchain technology, and digital payment mechanisms to establish a secure Farm-to-Consumer ecosystem. The frontend, developed using HTML, CSS, and JavaScript, enables farmers to upload authentic crop data and consumers to access transparent product information. The backend, implemented using Spring Boot, coordinates communication between system components, while the AI module developed using Python and Scikit-learn and deployed via FastAPI generates personalized meal recommendations from structured CSV datasets. Blockchain technology ensures immutable recording of transactions and supply-chain events through smart contracts, providing end-to-end traceability and data integrity. MySQL and NoSQL databases support structured and

unstructured data storage, while UPI enables fast and secure payments. Analytical evaluation indicates improved transparency, reduced intermediaries, secure transactions, and enhanced decision support compared to traditional agricultural systems, validating the effectiveness of the proposed model.

IV. RESULTS AND DISCUSSION

The experimental results of the FarmNexis+ platform demonstrate that the proposed system effectively enables direct interaction between farmers and consumers, significantly reducing dependency on intermediaries. The integration of Blockchain technology ensures end-to-end transparency and traceability, as every transaction and supply-chain event is securely stored in an immutable ledger. This improves trust, data integrity, and accountability compared to traditional centralized agricultural systems.

The AI-based meal recommendation system, developed using Python and Scikit-learn and deployed through FastAPI, successfully generates personalized meal suggestions based on selected vegetables. Although trained on a limited dataset, the system provides meaningful and nutrition-oriented recommendations, promoting informed food choices and reducing food wastage. Backend coordination through Spring Boot ensures smooth communication between frontend, AI services, blockchain modules, and databases.

The integration of UPI digital payments results in fast, secure, and low-cost transactions between farmers and consumers. Overall system evaluation indicates improved operational efficiency, secure data handling using MySQL and NoSQL databases, and reliable cloud deployment using Firebase, Render, and Docker. These results confirm that FarmNexis+ performs more efficiently than traditional agricultural models by enhancing transparency, transaction speed, and trust.

Sr. No.	Condition	Output
1	Farmer–Consumer direct interaction	Reduced intermediaries and fair pricing
2	Blockchain-based transaction recording	Improved transparency and data integrity
3	AI meal recommendation module	Personalized and nutrition-oriented suggestions
4	UPI payment integration	Fast, secure, and low-cost transactions
5	Cloud-based deployment with Docker	Scalable and reliable system performance

Table 1. Performance Result

V. CONCLUSION

In conclusion, FarmNexis+ successfully demonstrates how the integration of Artificial Intelligence, Blockchain technology, and digital payment systems can transform the traditional agricultural supply chain into a secure, transparent, and farmer-centric ecosystem. By enabling direct farm-to-consumer interaction, the platform reduces dependency on intermediaries and ensures fair pricing for farmers while providing consumers with trustworthy

and traceable products. The use of AI-based meal recommendation models deployed via FastAPI, along with a Spring Boot backend, enhances user experience and promotes nutritional awareness. Blockchain integration using smart contracts ensures end-to-end traceability, data integrity, and transparency across all supply-chain activities, while UPI-based payments enable fast, secure, and low-cost transactions. Supported by cloud deployment, MySQL, and NoSQL databases, the experimental results indicate improved efficiency, trust, and operational reliability compared to traditional agricultural models, validating FarmNexis+ as a practical and scalable solution for smart and sustainable digital agriculture.

ACKNOWLEDGEMENTS

We sincerely thank our project guide, Prof. A. V. Mahalle, and the Department of Computer Engineering, Jagadambha College of Engineering and Technology, Yavatmal, for their guidance and support.We extend our appreciation to all faculty members and peers for their constructive feedback and support.We also acknowledge the use of AI, Blockchain, FastAPI, Spring Boot, UPI, and cloud technologies, as described in this project, which enabled the successful development of the FarmNexis+ smart agriculture platform.

VI. REFERENCES

[1] Sriramoju, Sumanth, Mangesh Ingle, and Ashish Mahalle. "Mobile messaging application framework for service usage categorization." *International Journal of Research in Science and Engineering* 3.4 (2017): 56-65.

[2] Babu, Shoban, Mangesh Ingle, and Ashish Mahalle. "HLA based solution for packet loss detection in mobile ad hoc networks." *International Journal of Research In Science & Engineering* 3.4 (2017): 2394-8280.

[3] Jangid, Miss Komal T., Miss Santoshi R. Sharma, Miss Vrushali P. Chaudhari, Miss Samruddhi, K. Joshi, Miss Harsha V. Nikhade, and A. V. Mahalle. "ONLINE BANKING SYSTEM." *European Journal of Molecular & Clinical Medicine* 8, no. 04 (2021): 2021.

[4] Thakre, Swati Vijay, and Ashish V. Mahalle. "A Survey on Image Compression Techniques Using Artificial Neural Networks."

[5] Ms.Gauri M. Mankar ,Prof. Ashish V. Mahalle. "LATEST BUZZWORD OF CLOUD COMPUTING" *International Journal of Research In Science & Engineering e-ISSN: 2394-8299 Volume: 3 Issue: 5 Sep.-Oct. 2017.*

[6]Babu, S., Ingle, M., &Mahalle, A. TRUST AND ITERATIVE FILTERING APPROACHES FOR SECURE DATA COLLECTION IN WIRELESS SENSOR NETWORKS.

[7] D. Kundu, S. Kumar, Y. Sharma, S. Pathan, D. Theng and A. Mahale, "SPEEDZO: Vehicle speed detection using Video Processing," 2023 11th International Conference on Emerging Trends in Engineering & Technology - Signal and Information Processing (ICETET - SIP), Nagpur, India, 2023, pp. 1-6, doi: 10.1109/ICETET-SIP58143.2023.10151654.

[8] Mahalle, A. ., Khandelwal, S., Dhore, A. ., Barbudhe, V. ., & Waghmare, V. . (2024). Cyber attacks on UAV networks: A comprehensive survey. *Review of Computer Engineering Research*, *11*(1), 45–57. https://doi.org/10.18488/76.v11i1.3636.